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Variational Quantum Factorization (VQF) using Different Variations of Quantum Approximate Optimization Algorithm (QAOA)

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***Abstract*—Variational quantum factoring (VQF), a variational quantum algorithm, has been proposed as an alternative to Shor's algorithm for factoring numbers and has been experimentally implemented to factor numbers much larger than what Shor's algorithm has factorized thus far. The focus of this work is comparing the performance of different variations of quantum approximate optimization algorithm (QAOA) in terms of implementing VQF. Hence, VQF has been implemented here on the Qiskit quantum simulation framework using standard QAOA, multi-angle QAOA (maQAOA), conditional value at risk (CVaR) QAOA, and CVar maQAOA. Using these methods, VQF has been implemented for factorizing bi-prime numbers 143 and 291311 using 4 and 6 qubits respectively, and for factorizing a tri-prime number, 175, using 3 qubits. For factoring of 143 and 175, maQAOA yields the highest success probability numbers (0.92 and 0.99 respectively), while for factoring 291311, CVar maQAOA with 0.75 of the total measurements contributing to the expectation value yields the highest success probability.**

***Index Terms*—Quantum computing, QAOA, Quantum algorithms, NISQ devices**

I. Introduction

Cryptographic security currently relies heavily on the difficulty of factorizing large numbers, a fundamental concept underpinning classical encryption techniques. The process involves encoding a message using a large number in such a way that decryption necessitates knowledge of its factors. While it is extremely difficult for conventional classical (digital) computers to come up with these factors, one of the major advantages of quantum computers is considered to be the ability to calculate these factors with the help of Shor's algorithm [1], [2].

Theoretically, Shor's algorithm computes the factors of a number N in time which is a polynomial function of logarithm of N. On the other hand, the best classical approximation algorithm works in sub-exponential time to factorize the same number. However, implementing Shor's algorithm for a large enough number either needs a few noiseless qubits or a very large number of noisy qubits (to carry out error correction). But unfortunately, currently available noisy intermediate scale quantum (NISQ) hardware can offer neither [3].

Variational quantum algorithms, including Quantum Approximate Optimization Algorithm (QAOA), which are meant to be implemented on current NISQ hardware with 50-100 noisy qubits, have been proposed to approximately solve different kinds of machine learning and optimization problems [4], [5]. Variational Quantum Factorization (VQF) algorithm is one such variational algorithm which is meant to factorize large numbers on NISQ hardware. And while experimental implementation of Shor's algorithm has been limited to factoring numbers as small as 21 and 35 thus far [6], [7], VQF has been experimentally implemented to factorize larger numbers like 55, 65, 77, and 91 [8] and also numbers as large as 3127, 6557, and 1099551473989 [9] given the higher robustness of VQF to noise compared to that of Shor's algorithm.

The focus of this paper is comparative analysis of different variations of QAOA with respect to solving the VQF problem. Hence, we solve VQF here for two bi-prime numbers (143 and 291311) and one tri-prime number (175) using four variations of QAOA: standard QAOA (Fig. 1), multi-angle QAOA (maQAOA), conditional value at risk QAOA (CVar QAOA), and a combination of maQAOA and CVar QAOA. We use Qiskit quantum simulation package for this purpose.

Previous studies have explored VQF using standard QAOA. On the other hand, various QAOA variants have been successfully applied to solve a wide range of other combinatorial optimization problems (but not including VQF) [10]–[13]. To the best of our knowledge, our paper here is the first report that applies and compares multiple QAOA variants in the context of VQF. We show that out of all the variations we have explored, maQAOA and CVar variation of maQAOA perform the best for all three numbers we have explored here.

In Section II, we describe the method we have followed to implement VQF and different variations of QAOA while providing relevant equations. In Section III, we discuss our obtained results. In Section IV, we conclude the paper.

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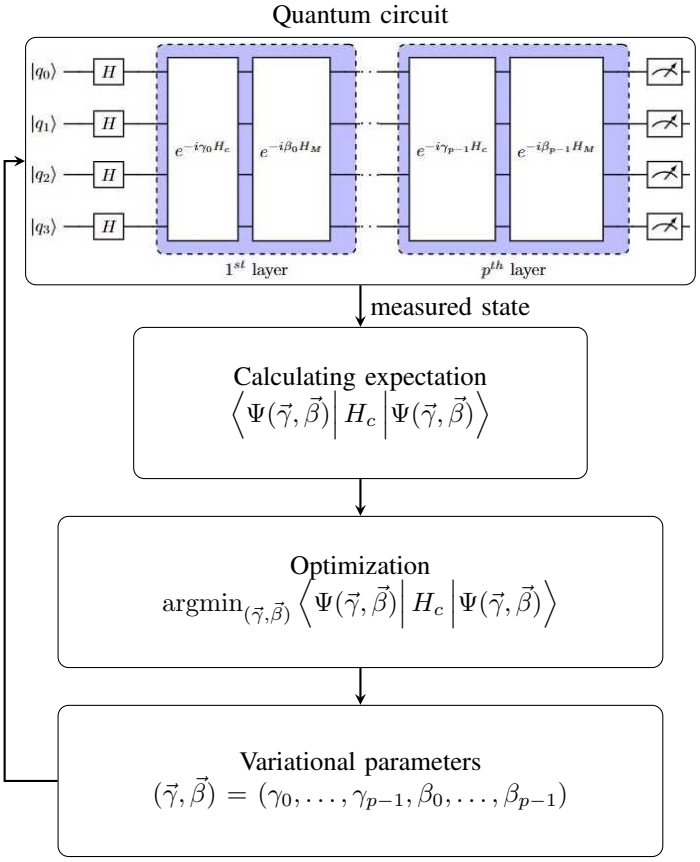

Fig. 1: Schematic showing a QAOA circuit with p stages and the method for optimizing its parameters (more details in text)

II. Theory and Methods

A. Variational Quantum Factorization (VQF)

Following the method introduced by Burges [14], the factorization problem can be transformed into an optimization problem as follows. For a bi-prime number m, where $m = p \times q$ (p and q being prime numbers), these prime factors p and q need to be evaluated, which can be seen as the reverse process of traditional binary multiplication.

To illustrate this, let's consider the factorization of the bi-prime number 143, where p and q are represented as 4-bit numbers $1p_2p_11$ and $1q_2q_11$, respectively. To explore a nontrivial case, we set m to be odd, ensuring the least significant bit of the multipliers is 1. The bit width of m must equal the sum of the bit widths of the two multipliers. Here, we focus on a specific case where both multipliers have equal bit widths, with the most significant bit set to 1. In Table I [15], in the first two rows p_i and q_i are the individual bits of the multipliers. The following four rows exhibit the intermediate outcomes of the multiplication process, where $z_{i,j}$ denotes the carryovers from the ith bit to the jth bit.

With respect to Table I, after adding the contents of each column, the following clauses are obtained:

	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
p					1	p_1	p_0	1
q					1	q_1	q_0	1
					1	p_1	p_0	1
				q_0	p_1q_0	p_0q_0	q_0	$\times$
			q_1	p_1q_1	p_0q_1	q_1	$\times$	$\times$
		1	p_1	p_0	1	$\times$	$\times$	$\times$
carries	$z_{5,6}$	$z_{4,5}$	$z_{3,4}$	$z_{2,3}$	$z_{1,2}$	$z_{0,1}$		
	$z_{4,6}$	$z_{3,5}$	$z_{2,4}$	$z_{1,3}$				
$p \times q = m$	1	0	0	0	1	1	1	1

TABLE I: Binary multiplication table for bi-prime number 143.

$$p_0 + q_0 = 1 + 2z_{0,1} \tag{1a}$$
$$p_1 + p_0q_0 + q_1 + z_{0,1} = 1 + 2z_{1,2} + 4z_{1,3} \tag{1b}$$
$$1 + p_1q_0 + p_0q_1 + 1 + z_{1,2} = 1 + 2z_{2,3} + 4z_{2,4} \tag{1c}$$
$$q_0 + p_1q_1 + p_0 + z_{1,3} + z_{2,3} = 1 + 2z_{3,4} + 4z_{3,5} \tag{1d}$$
$$q_1 + p_1 + z_{3,4} + z_{2,4} = 0 + 2z_{4,5} + 4z_{4,6} \tag{1e}$$
$$1 + z_{4,5} + z_{3,5} = 0 + 2z_{5,6} \tag{1f}$$
$$z_{5,6} + z_{4,6} = 1. \tag{1g}$$

We next apply the rules for simplification outlined by Anschuetz *et al* [16]:

$$xy - 1 = 0 \implies x = y = 1 \tag{2a}$$
$$x + y - 1 = 0 \implies xy = 0 \tag{2b}$$
$$a = bx = 0 \implies x = 1 \tag{2c}$$
$$\sum x_i = 0 \implies x_i = 0 \forall i \tag{2d}$$
$$\sum_{i=1}^{a} x_i - a = 0 \implies x_i = 1 \forall i, \tag{2e}$$

where $x, y, x_i \in \{0, 1\}$ be unknown binary variables, and $a, b \in \mathbb{Z}^+$ are the positive constants. Using the above simplification rules, the clauses for factorization of 143 (equation 1) can be reduced to:

$$p_0 + q_0 - 1 = 0 \tag{3a}$$
$$p_1 + q_1 - 1 = 0 \tag{3b}$$
$$p_0q_1 + q_0p_1 - 1 = 0. \tag{3c}$$

Let the equations (3a), (3b), and (3c) be denoted as clauses C_i's (i =1, 2, 3). Since the square of any real number is always greater than or equal to zero, the minimum value of the function $(p_0 + q_0 - 1)^2$ is 0. Hence, the squares of all the simplified clauses can be added and the sum can be equated to 0 in order to calculate the values for p_0, p_1, q_0 and q_1. Thus, the objective function corresponding to factorization of 143 can be written as:

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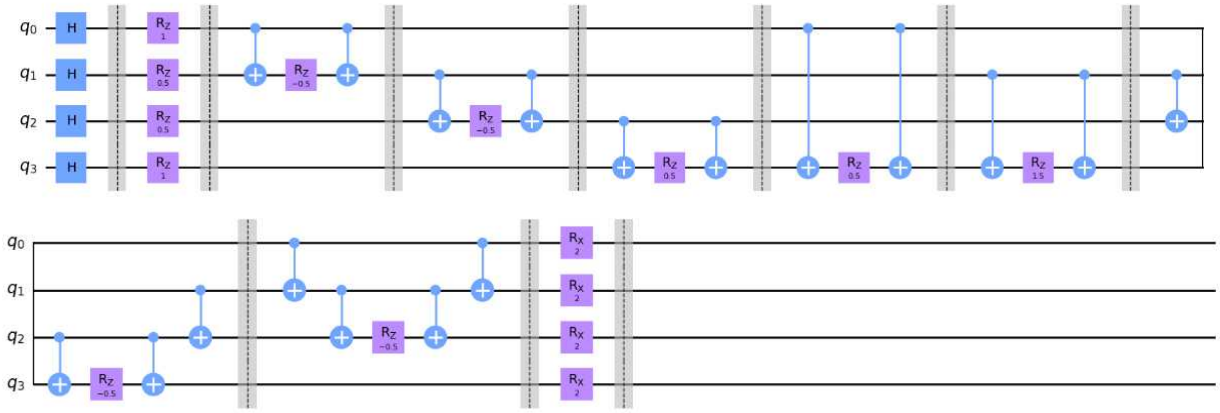

Fig. 2: QAOAansatz/ circuit for factoring 143, as designed on Qiskit simulation framework

$$f(x) = \sum_{i=1}^{3} \hat{C}_i^2 = (p_0 + q_0 - 1)^2 + (p_1 + q_1 - 1)^2 + (p_1 q_0 + p_0 q_1 - 1)^2, \quad (4)$$

where $x \in \{0,1\}^n$ corresponding to the bitstring formed by the variables p_0, p_1, q_0, and q_1, and n is the number of clauses (n=4 for 143). However, this method of construction leads to a four-body interaction term in $f(x)$, which poses challenges for experimental implementation. Following the alternative approach formulated by Schaller *et al* [17], the objective function can be reformulated to only contain upto 3-body interactions [18] and hence can be written as:

$$\bar{f}(x) = 5 - 3p_0 - p_1 - q_0 + 2p_0q_0 - 3p_1q_0 + 2p_0p_1q_0 - 3q_1 + p_0q_1 + 2p_1q_1 + 2p_1q_0q_1, \quad (5)$$

which can assume a minimum value of 0.

The classical objective function $\bar{f}(x)$, which has to be minimized, is further reformulated into a more suitable form for quantum optimization [19]. This reformulation is done by replacing each classical bit value $a_k \in \{p_i, q_i\}$ by a corresponding qubit operator using the transformation:

$$a_k \rightarrow \frac{I - \sigma_k^z}{2}, \quad (6)$$

where σ_k^z is the Pauli-z operator acting on the kth qubit. This mapping ensures that when operated on qubits $|0\rangle$ and $|1\rangle$, the resulting values align with the classical bit assignments $a_k = 0$ and $a_k = 1$, respectively. Thus, the classical objective function $\bar{f}(x)$ is converted into an Ising Hamiltonian in the quantum domain and encoded in terms of a quantum circuit, where the interaction terms are taken care of as multi-qubit gates (R_{ZZ} etc.). Solving the factorization problem is thereby same as finding the ground state of the Ising Hamiltonian, which is achieved using QAOA and its variations, as described below.

B. Quantum Approximate Optimization Algorithm (QAOA) and its variations: maQAOA, CVar QAOA, CVar maQAOA

The factorization problem's objective function, as provided in equation 5, is reformulated as a Hamiltonian H_c. Thus, the problem Hamiltonian for 143 is:

$$H_c = -\frac{\sigma_0^z\sigma_1^z\sigma_2^z}{4} + \frac{\sigma_0^z\sigma_1^z}{4} + \frac{3\sigma_0^z\sigma_2^z}{4} + \frac{\sigma_0^z\sigma_3^z}{4} + \frac{\sigma_0^z}{2} - \frac{\sigma_1^z\sigma_2^z\sigma_3^z}{4} - \frac{\sigma_1^z\sigma_2^z}{4} + \frac{3\sigma_1^z\sigma_3^z}{4} + \frac{\sigma_1^z}{4} + \frac{\sigma_2^z\sigma_3^z}{4} + \frac{\sigma_2^z}{4} + \frac{\sigma_3^z}{2} + 2 \quad (7)$$

For every eigenstate $|u_x\rangle$,

$$H_c |u_x\rangle = f(x) |u_x\rangle . \quad (8)$$

The eigenstate $|u_x\rangle$ corresponding to the lowest eigenvalue of H_c is the solution to the VQF problem, and QAOA aims to find this eigenstate $|u_x\rangle$.

As shown in Fig. 1, the QAOA ansatz (quantum circuit) comprises a sequence of p layers, each featuring two unitary operators, $e^{-i\beta H_M}$ and $e^{-i\gamma H_c}$, where H_M is the mixer Hamiltonian:

$$H_M = -\sum_{j=0}^{n-1} \sigma_j^x, \quad (9)$$

and H_c is the problem Hamiltonian:

$$H_c = \sum_{\alpha \in \{0,1\}^n} h_\alpha \otimes_{i=1}^{n} \sigma_i^{\alpha_i}, \quad (10)$$

where $h_\alpha \in \mathbb{R}, \quad \forall \alpha \in \{0,1\}^n$

Plots for success probability as a function of number of stages/ layers (p) in the quantum circuit for factoring bi-prime number 143 using different methods: standard QAOA (a) and maQAOA (a), CVar QAOA (b), and CVar maQAOA (c). Within the standard QAOA formalism, after initializing the qubits at an initial superposition state $\frac{|0\rangle+|1\rangle}{2}^{\otimes n}$ (with the help of Hadamard gates), the two aforementioned unitary operators are operated alternatively p times (number of stages = p) as shown below:

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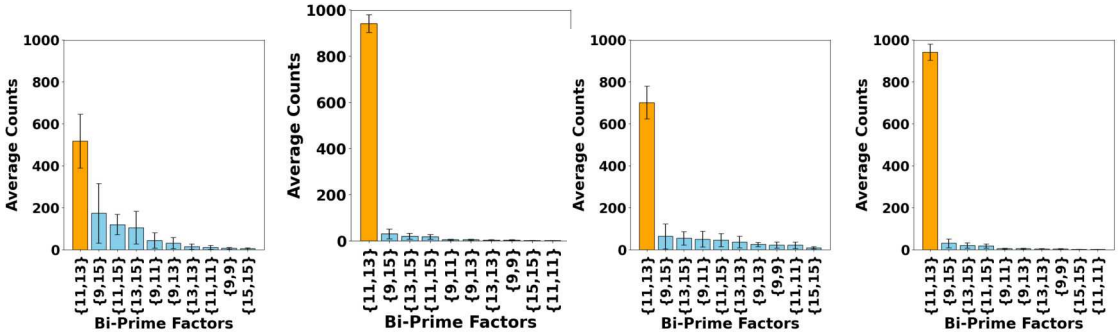
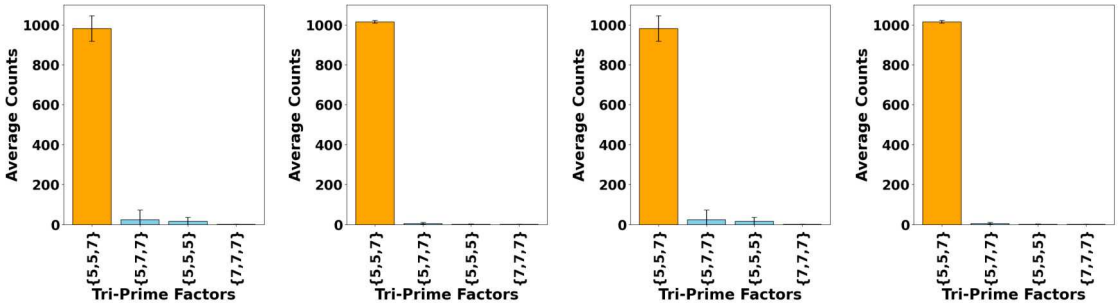


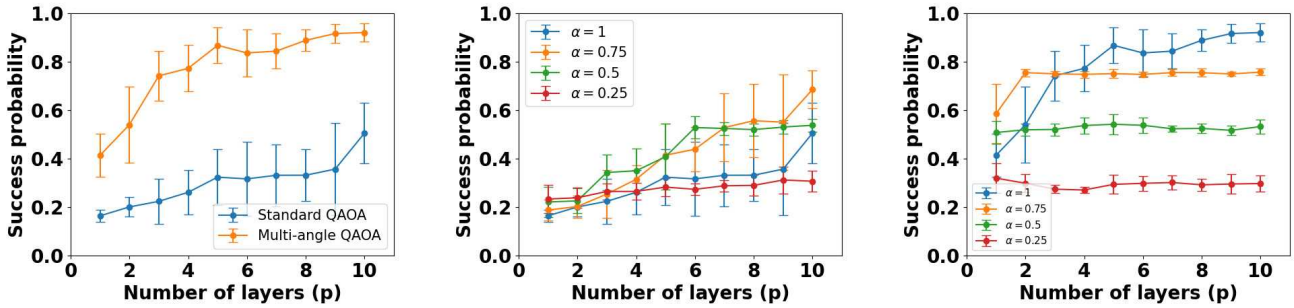
(a) Standard QAOA (b) Multi-angle QAOA (c) Cvar QAOA (d) Multi-angle Cvar QAOA

Fig. 3: Measurement statistics/ frequency plots for factoring bi-prime number 143 for all variants of QAOA. 11 and 13 are the correct factors, and they occur with the highest probability for maQAOA (b) and CVar maQAOA with $\alpha = 1$, which is same as maQAOA (d).



(a) Standard QAOA (b) Multi-angle QAOA (c) Cvar QAOA (d) Multi-angle Cvar QAOA

Fig. 4: Measurement statistics/ frequency plots for factoring tri-prime number 175 for all variants of QAOA. 5, 5, and 7 are the correct factors, and they occur with the highest probability for maQAOA (b) and CVar maQAOA with $\alpha = 1$, which is same as maQAOA (d).



(a) Standard and Multi-angle QAOA (b) CVar QAOA (c) Multi-angle Cvar QAOA

Fig. 5: Plots for success probability as a function of number of stages/ layers (p) in the quantum circuit for factoring bi-prime number 143 using different methods: standard QAOA (a) and maQAOA (a), CVar QAOA (b), and CVar maQAOA (c). Different α values for CVar QAOA and CVar maQAOA are listed in the legends of (b) and (c). α is defined with respect to the CVar technique in equation 13.

$$\left|\Psi_p(\vec{\gamma},\vec{\beta})\right\rangle = e^{-i\beta_{p-1}H_M}e^{-i\gamma_{p-1}H_c}\dots e^{-i\beta_0 H_M}e^{-i\gamma_0 H_c}\left|+\right\rangle^{\otimes n}. \tag{11}$$

Thus, in standard QAOA, there are total p variational parameters $\vec{\gamma} = (\gamma_0, \dots, \gamma_{p-1})$, for $\gamma_i \in [0, 2\pi)$ and p variational parameters $\vec{\beta} = (\beta_0, \dots, \beta_{p-1})$ for $\beta_i \in [0, \pi)$. Hence, in standard QAOA, for a given stage, all gates corresponding to the mixer Hamiltonian use the same parameter β, and all gates corresponding to the problem Hamiltonian use the same parameter γ.

However, in multi-angle QAOA (maQAOA) [13], at every stage of the quantum circuit, every gate corresponding to the mixer Hamiltonian H_M has a different parameter (β_1, β_2, β_3, etc.), and every gate corresponding to the problem Hamiltonian H_c has a different parameter ($\gamma_{1,2}$ for a two-

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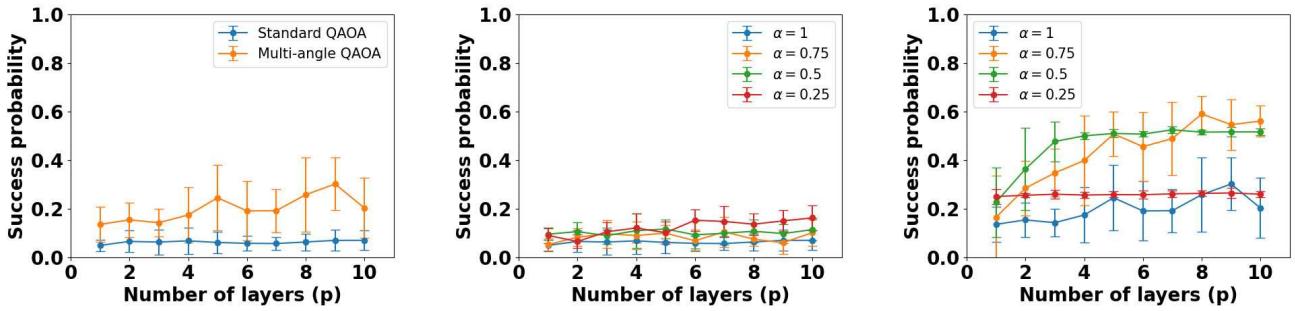
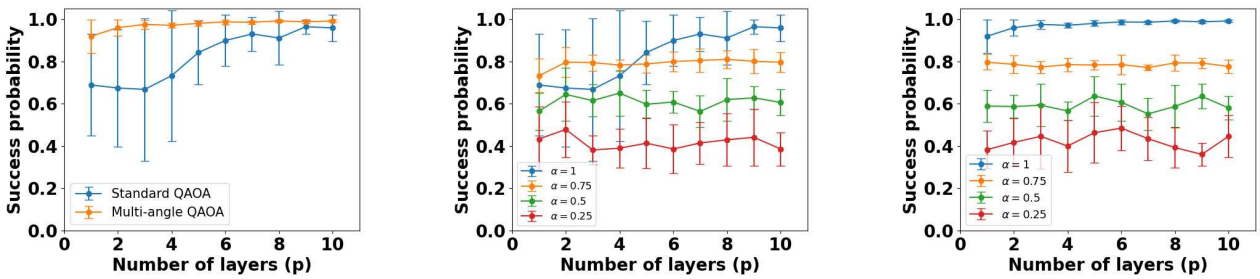


(a) Standard and Multi-angle QAOA (b) CVar QAOA (c) Mutli-angle Cvar QAOA

Fig. 6: Plots for success probability as a function of number of stages/ layers (p) in the quantum circuit for factoring bi-prime number 291311 using different methods: standard QAOA (a) and maQAOA (a), CVar QAOA (b), and CVar maQAOA (c). α is defined with respect to the CVar technique in equation 13.



(a) Standard and Multi-angle QAOA (b) CVar QAOA (c) Multi-angle Cvar QAOA

Fig. 7: Plots for success probability as a function of number of stages/ layers (p) in the quantum circuit for factoring tri-prime number 175 using different methods: standard QAOA (a) and maQAOA (a), CVar QAOA (b), and CVar maQAOA (c). α is defined with respect to the CVar technique in equation 13.

qubit gate connecting qubits 1 and 2, $\gamma_{2,3}$ for a two-qubit gate connecting qubits 2 and 3, etc.). Thus maQAOA has several more parameters to optimize across compared to QAOA.

In either case, a classical optimization algorithm is utilized in feedback (Fig. 1) to determine the optimal values of $\vec{\gamma}$ and $\vec{\beta}$ by minimizing the expectation value:

$$E(\vec{\gamma},\vec{\beta}) = \langle \vec{\gamma},\vec{\beta}|H_c|\vec{\gamma},\vec{\beta}\rangle \tag{12}$$

of the problem Hamiltonian H_c, which is effectively same as finding the ground state of the Hamiltonian through energy minimization.

As proposed by Barkoutsos *et al* [12], the basic premise of the conditional value at risk (CVar) technique is that calculating expectation value, the $g(\theta) = \frac{1}{N}\sum_{i=1}^{N} H_c(x_i)$ (where $N \in \mathbb{N}$ is the total number of samples/ measurements of the quantum state and and each bit string x represents a specific outcome of the quantum measurement) may not be the most practical choice. It is more reasonable to enhance the properties of the best measurement outcome rather than focusing on the average result. Hence, in CVar QAOA and CVar ma-QAOA, all possible n-bit strings are arranged in the increasing order, and only a fraction of measurements with the highest measurement outcomes ($N\alpha$, where $\alpha < 1$) is considered to calculate the expectation value:

$$g_\alpha(\theta) = \frac{1}{N\alpha}\sum_{i=1}^{N\alpha} H_c(x_i). \tag{13}$$

III. Results

Based on the method described in Section II, we have implemented QAOA and its variants on Qiskit simulation framework [20]. The design of the quantum circuit for factoring 143 is discussed next (4 qubits needed).

$$\begin{aligned}
U_c =& \text{Exp}(H_c,\gamma) = e^{-iH_c\gamma} && \text{(14a)}\\
=& \text{Exp}\Big(-\frac{\sigma_0^z\sigma_1^z\sigma_2^z}{4} + \frac{\sigma_0^z\sigma_1^z}{4} + \frac{3\sigma_0^z\sigma_2^z}{4} + \frac{\sigma_0^z\sigma_3^z}{4} + \frac{\sigma_0^z}{2} - \\
& \frac{\sigma_1^z\sigma_2^z\sigma_3^z}{4} - \frac{\sigma_1^z\sigma_2^z}{4} + \frac{3\sigma_1^z\sigma_3^z}{4} + \frac{\sigma_1^z}{4} + \frac{\sigma_2^z\sigma_3^z}{4} + \frac{\sigma_2^z}{4} + \\
& \frac{\sigma_3^z}{2} + 2, \gamma\Big) && \text{(14b)}\\
=& R_{012}^{zzz}(\frac{-\gamma}{4})R_{01}^{zz}(\frac{\gamma}{4})R_{02}^{zz}(\frac{3\gamma}{4})R_{03}^{zz}(\frac{\gamma}{4})R_0^z(\frac{\gamma}{2})R_{123}^{zzz}(-\frac{\gamma}{4}) \\
& R_{01}^{zz}(-\frac{\gamma}{4})R_{13}^{zz}(\frac{3\gamma}{4})R_1^z(\frac{\gamma}{4})R_{23}^{zz}(\frac{\gamma}{4})R_2^z(\frac{\gamma}{4})R_3^z(\frac{\gamma}{4}). && \text{(14c)}
\end{aligned}$$

Similarly for U_M we get,

$$\begin{aligned}
U_M =& \text{Exp}(H_M,\beta) = e^{-iH_M\beta} \\
=& \text{Exp}\Big(-\sum_{j=0}^{n-1}\sigma_j^x, \beta\Big) \\
=& \prod_{j=0}^{3} R_x(2\beta).
\end{aligned} \tag{15}$$

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This decomposition of the unitary evolutions of the Hamiltonians into various gates leads to the quantum circuit shown in Fig. 2. Similar decomposition leads to 4-qubit-based circuit design for 291311 and 3-qubit-based circuit design for 175.

For a given number, we have obtained the obtained factors for every measurement and have counted the number of measurements for which the product of the factors is equal to the given number (number of successful trials). Such measurement results for factoring 143 and 175 are shown in Fig. 3 and Fig. 4 respectively.

Fig. 5, 6, and 7 provide plots of obtained success probability as a function of number of layers (p) for bi-prime number 143, bi-prime number 291311, and tri-prime number 175 respectively for standard QAOA (sub-plot (a)), ma QAOA (sub-plot (a), CVar QAOA (sub-plot (b)), and CVar ma-QAOA (sub-plot (c)). As the number of layers increases, success probability increases (which is desired) for some QAOA techniques for some particular numbers but doesn't increase for some other numbers. Thus, we find no technique that works uniformly for all numbers.

Nonetheless, maQAOA and CVar maQAOA (for some α values) on an average yield the highest success probability numbers among all the variants. For example, the highest success probability of 0.92 has been obtained for factoring 143 using maQAOA technique (or $\alpha = 1$ in CVar-maQAOA technique which makes it same as maQAOA). This is consistent with the measurement statistics / frequency plots shown in Fig. 3. Highest success probability of 0.56 has been obtained for factoring 291311 using CVar maQAOA technique with α =0.75. Highest success probability of 0.99 has been obtained for factoring 175 using maQAOA technique (or CVar maQAOA technique with $\alpha = 1$), consistent with the frequency plots shown in Fig. 4. These numbers have been listed in Table II.

As per our results, among the three numbers we have explored, the number 175 can be factored with the highest success probability cutting across different methods. This can be attributed to the lower circuit complexity and gate count associated with factoring 175 compared to 143 and 291311. We observe that factoring 291311 has the lowest success probability since it involves largest problem Hamiltonian and hence the most complex quantum circuit. Higher the gate noise and circuit complexity, poorer is the success probability.

When compared with other reports, no other paper compares different variations of QAOA for implementing VQF unlike this paper, as mentioned earlier. Even with respect to standard QAOA for VQF, in reports by Selavarajan *et al* [8] and by Karamlou *et al* [9], VQF has been implemented experimentally for numbers that are different from that used in this paper. In the report by Anschuetz *et al* [16], VQF for factoring 291311 has been simulated, but success probability has not been plotted making it hard to compare with our results.

Number factorized	Number of qubits	Highest success probability	QAOA variant used
143	4	0.92	Multi angle QAOA; Multi angle CVar QAOA (α =1)
291311	6	0.56	Multi angle CVar QAOA (α =0.75)
175	3	0.99	Multi angle QAOA Multi angle CVar QAOA (α =1)

TABLE II: Results summary: Numbers factorized, number of qubits needed, highest success probability achieved, and variant of QAOA used to obtain this success probability. α is defined with respect to the CVar technique in equation 13.

IV. Conclusion

In summary, to factor large numbers, we have here implemented a variational algorithm, VQF, here on the Qiskit quantum simulation framework using standard QAOA, multi-angle QAOA (maQAOA), conditional value at risk (CVaR) QAOA, and CVar maQAOA. Factoring of two bi-prime numbers and one tri-prime number has been explored using these methods. Multi-angle QAOA and CVar variation of multi-angle QAOA have been found to offer the best performance among all the explored methods. Our work paves the way for factorizing large numbers (one of the key applications that drive quantum computing research) on NISQ hardware, which is not suitable for implementing Shor's algorithm to factor the numbers.

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